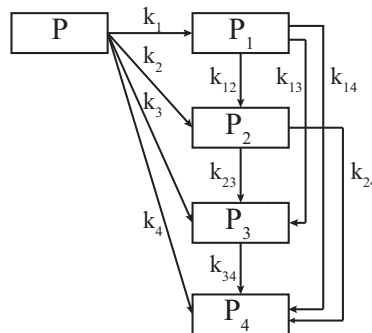


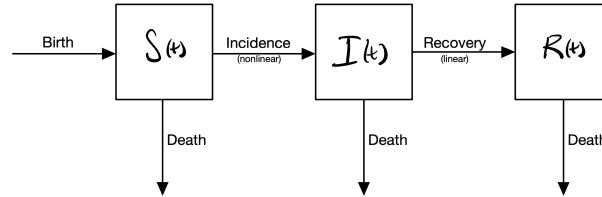
HIGHER DIMENSIONAL SYSTEMS

- 1.] The following schematic represents a chemical process in which a parent peptide (denoted by P) breaks down into four smaller components (denoted by P_i for $i = 1, 2, 3, 4$). The larger components can also break down into smaller components, where P_4 is the smallest component. Here, assume each “bin” is a function of time and measures the concentration of each component in the solution.



Assume that the initial condition contains only the parent peptide, $P(0) = 1$, and all other components are not yet formed: $P_i(0) = 0$ for $i = 1, 2, 3, 4$. Set up and write down the linear differential equation model that simulates this process. Assume each reaction is constant and positive. Create a file that solves this model and outputs the solution when $k_1 = .09$, $k_2 = .8$, $k_3 = .07$, $k_4 = .6$, $k_{12} = .05$, $k_{13} = .5$, $k_{14} = .06$, $k_{23} = .7$, $k_{24} = .08$, and $k_{34} = .9$.

- 2.] An SIR model is a system of three differential equations that models the spread of an *infectious disease* throughout a population. Consider the compartment based model below and the associated system of equations. The classic endemic SIR model can be described by the following diagram:



Here, $S(t)$, $I(t)$, and $R(t)$ are the fraction of the total population that are susceptible, infected, or recovered, respectively. We can assemble the following equations based on this diagram:

$$\begin{aligned}\frac{dS}{dt} &= \alpha - \beta SI - \mu S \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I \\ \frac{dR}{dt} &= \gamma I - \mu R\end{aligned}$$

Solve this in Matlab and plot a trajectory as a time-series and on the three-dimensional phase diagram.

- 3.] The *Lorenz system* is a three-dimensional nonlinear system of differential equations originally developed by a meteorologist as a simplified mathematical model of atmospheric convection. The phase portrait has essentially become a symbol for deterministic chaos. Here it is:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

Solve this in Matlab and plot a trajectory as a time-series and on the three-dimensional phase diagram with $\sigma = 10$, $\beta = 8/3$, and $\rho = 28$.